

### 0.0.1 Locally Compact

**Definition 0.0.1.** A Space  $X$  is called *Locally Compact* if:

For any  $x \in X$ , there exist open  $U$  and compact  $C$  such that  $x \in U \subseteq C$ .

### 0.0.2 One-point Compactification

**Definition 0.0.2.** Let  $(X, \mathcal{T})$  be a Space.

Define  $X_\infty \stackrel{\text{def}}{=} X \sqcup \{\infty\}$  and  $\mathcal{T}_\infty \stackrel{\text{def}}{=} \mathcal{T} \sqcup \{U \subseteq X_\infty \mid \infty \in U, X_\infty \setminus U \text{ is compact in } X\}$ .

This  $(X_\infty, \mathcal{T}_\infty)$  is called *one-point compactification* of  $X$ .

**Theorem 0.0.3.** Let  $(X, \infty)$  be a Locally-Compact Hausdorff Space, but not Compact.

Then, one-point compactification  $(X_\infty, \mathcal{T}_\infty)$  of  $X$  is Compact Hausdorff Space.

*Proof.* This proof consisted of five steps.

1). Claim:  $\mathcal{T}_\infty$  is Topology on  $X_\infty$ . (Using  $X$  is Hausdorff)

Let  $U_\gamma \in \Gamma$ ,  $(\gamma \in \Gamma)$  be elements of  $\mathcal{T}_\infty$ .

Define  $\Gamma_1 \stackrel{\text{def}}{=} \{\alpha \in \Gamma \mid U_\alpha \in \mathcal{T}\}$ , and  $\Gamma_2 \stackrel{\text{def}}{=} \Gamma \setminus \Gamma_1 = \{\beta \in \Gamma \mid \infty \in U_\beta, X_\infty \setminus U_\beta \text{ is compact in } X\}$ .

Then,  $\bigcup_{\gamma \in \Gamma} U_\gamma = \left( \bigcup_{\alpha \in \Gamma_1} U_\alpha \right) \cup \left( \bigcup_{\beta \in \Gamma_2} U_\beta \right)$ . The left term is open in  $X$  clearly.

And, put  $C_\beta = X_\infty \setminus U_\beta$  for each  $\beta \in \Gamma_2$ . Then,  $C_\beta$  is Compact in  $X$  by definition, thus closed by  $X$  is Hausdorff.

$$\bigcup_{\beta \in \Gamma_2} U_\beta = \bigcup_{\beta \in \Gamma_2} X_\infty \setminus C_\beta = X_\infty \setminus \left( \bigcap_{\beta \in \Gamma_2} C_\beta \right)$$

This intersection of  $C_\beta$  is compact, being any intersection of closed is closed and closed subset of compact.

That is, it is compact in  $X$ , therefore this union of  $U_\beta$  is contained in  $\mathcal{T}_\infty$ .

Let  $U_1, U_2 \in \mathcal{T}$ , and  $V_1, V_2 \in \mathcal{T}_\infty \setminus \mathcal{T}$ . Put  $C_i \stackrel{\text{def}}{=} X_\infty \setminus V_i$ ,  $(i = 1, 2)$ . Then,  $C_i$  is compact. Now,

$$\begin{aligned} U_1 \cap U_2 &\in \mathcal{T} \subset \mathcal{T}_\infty \\ U_1 \cap V_1 &= U_1 \cap (X_\infty \setminus C_1) = U_1 \cap X_\infty \cap C_1^c = U_1 \cap C_1^c = U_1 \setminus C_1 \in \mathcal{T} \subset \mathcal{T}_\infty \\ V_1 \cap V_2 &= (X_\infty \setminus C_1) \cap (X_\infty \setminus C_2) = X_\infty \setminus (C_1 \cap C_2) \in \mathcal{T}_\infty \end{aligned}$$

Thus closed under the arbitrary union and finite intersection.

2). Claim:  $(X, \mathcal{T})$  is a Subspace of  $(X_\infty, \mathcal{T}_\infty)$ . That is,  $\mathcal{T} = \{U \cap X \mid U \in \mathcal{T}_\infty\}$ . (Using  $X$  is Hausdorff)

The right inclusion is clear:  $U \in \mathcal{T} \implies U \in \mathcal{T}_\infty$ . Thus  $U = X \cap U \in \{U \cap X \mid U \in \mathcal{T}_\infty\}$ .

To show the left inclusion: Let  $U \in \mathcal{T}_\infty$ . If  $U \in \mathcal{T}$ , then  $X \cap U = U \in \mathcal{T}$ .

If  $U \notin \mathcal{T}$ , then  $X_\infty \setminus U$  is compact in  $X$ . Now,  $X \cap U = X \setminus \underbrace{(X_\infty \setminus U)}_{\text{compact in } T_2 \implies \text{closed}} \in \mathcal{T}$ .

3). Claim:  $\overline{X} = X_\infty$ . That is, closure of  $X$  is  $X_\infty$ . (Using  $X$  is not compact)

Let  $U \in \mathcal{T}_\infty$  with  $\infty \in U$ . Then,  $X_\infty \setminus U$  is compact of  $X$ , thus  $X_\infty \setminus U \subsetneq X$  because  $X$  is not compact.

4). Claim:  $X_\infty$  is Compact Space.

Let  $\mathcal{O} = \{U_\alpha \mid \alpha \in \Lambda\}$  be an open cover of  $X_\infty$ . Since  $\infty \in X_\infty = \bigcup_{\alpha \in \Lambda} U_\alpha$ , there is  $\alpha_0 \in \Lambda$  such that  $\infty \in U_{\alpha_0}$ .

$C \stackrel{\text{def}}{=} X_\infty \setminus U_{\alpha_0}$  is compact in  $X$ , thus so in  $X_\infty$ . And,  $C \subseteq \bigcup_{\alpha \in \Lambda \setminus \{\alpha_0\}} U_\alpha$ , thus there is finite subcover of  $C$ .

Finally, union of finite subcover of  $C$  and  $U_{\alpha_0}$  is finite subcover of  $X_\infty$ .

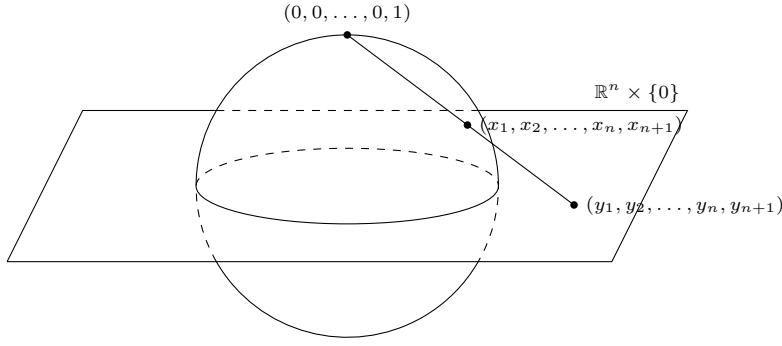
5). Claim:  $X_\infty$  is Hausdorff. (Using  $X$  is Locally-Compact)

Let  $x, y \in X_\infty$ . If both  $x, y$  are contained  $X$ , then there is nothing to prove, being  $X$  is hausdorff.

If  $x \in X$  and  $y = \infty$ , then there is open  $U$  and compact  $C$  of  $X$  such that  $x \in U \subseteq C$ , by Locaaly-Compact.

Now,  $x \in U$  and  $\infty \in X_\infty \setminus C$ , both are open of  $X_\infty$  with  $U \cap (X_\infty \setminus C) = \emptyset$ . □

### 0.0.3 Stereographic projection



**Definition 0.0.4.** Let  $\mathbb{R}^{n+1}$  be a Euclidean Space. Define *Unit Sphere* on  $\mathbb{R}^{n+1}$ :

$$S^n \stackrel{\text{def}}{=} \{(x_1, x_2, \dots, x_n, x_{n+1}) \in \mathbb{R}^{n+1} \mid \|(x_1, \dots, x_{n+1})\| = 1\}$$

**Theorem 0.0.5.** For any  $p \in S^n$ ,  $S^n \setminus \{p\} \cong \mathbb{R}^n$ .

*Proof.* WLOG, put  $p = (0, 0, \dots, 0, 1)$ . Define a map *Stereographic Projection*:

$$f : S^n \setminus \{p\} \rightarrow \mathbb{R}^n : (x_1, x_2, \dots, x_n, x_{n+1}) \mapsto \left( \frac{x_1}{1 - x_{n+1}}, \frac{x_2}{1 - x_{n+1}}, \dots, \frac{x_n}{1 - x_{n+1}} \right)$$

This map is derived by:

$$(y_1, y_2, \dots, y_n, y_{n+1}) = t(\vec{x} - \vec{p}) + (0, 0, \dots, 0, 1) = (x_1 t, x_2 t, \dots, x_n t, (x_{n+1} - 1)t + 1)$$

$y_{n+1} = 0$  when  $t = \frac{1}{1 - x_{n+1}}$ . Each  $(x_i, x_{n+1}) \mapsto \frac{x_i}{1 - x_{n+1}}$  is Continuous map, so  $f$  is Continuous map.

And, the map

$$g : \mathbb{R}^n \rightarrow S^n \setminus \{p\} : (x_i)_{i=1}^n \mapsto \left( \frac{2x_1}{1 + \|(x_i)\|^2}, \frac{2x_2}{1 + \|(x_i)\|^2}, \dots, \frac{2x_n}{1 + \|(x_i)\|^2}, 1 - \frac{2}{1 + \|(x_i)\|^2} \right)$$

is Continuous, and  $g = f^{-1}$ . Thus,  $f$  is Homeomorphism. □

**Theorem 0.0.6.** One Point Compactification of  $\mathbb{R}^n$  is  $S^n \subseteq \mathbb{R}^{n+1}$ .

*Proof.* Since  $\mathbb{R}$  is Locally-Compact Hausdorff Space, finite product Space  $\mathbb{R}^n$  is Locally-Compact Hausdorff Space. Let  $\mathbb{R}_\infty^n = \mathbb{R}^n \cup \{\infty\}$  be a One Point Compactification of  $\mathbb{R}^n$ . Then,  $\mathbb{R}_\infty^n$  is Compact Hausdorff Space. Meanwhile, put  $p = (0, 0, \dots, 0, 1)$ . Since  $S^n \setminus \{p\} \cong \mathbb{R}^n$ , there exists a Homeomorphism  $f : S^n \setminus \{p\} \rightarrow \mathbb{R}$ . Define

$$\tilde{f} : S^n \rightarrow \mathbb{R}_\infty^n : \begin{cases} f(x) & x \neq p \\ \infty & x = p \end{cases}$$

Then,  $\tilde{f}$  is Bijective map, and Continuous because: for any open  $U \subseteq \mathbb{R}_\infty^n$  containing  $\infty$ ,

$$\tilde{f}^{-1}[\mathbb{R}_\infty^n \setminus U] = f^{-1}[\mathbb{R}^n \setminus U]$$

Since  $\mathbb{R}^n \setminus U$  is Compact and  $f^{-1}$  is continuous,  $\tilde{f}^{-1}[\mathbb{R}_\infty^n \setminus U] = f^{-1}[\mathbb{R}^n \setminus U]$  is compact in  $S^n$ , thus closed. Moreover, since  $S^n$  is Compact and  $\mathbb{R}_\infty^n$  is Hausdorff,  $\tilde{f}$  is Closed map. Consequently,  $\tilde{f}$  is Homeomorphism. □